AMOXICILLIN

Ask about penicillin allergy

Pregnancy: compatible

CHEMICAL NAME (2S,5R,6R)-6-[[(2R)-2-amino-2-(4-hydroxyphenyl)acetyl]amino]-3,3-dimethyl-7-oxo-4-thia-1-azabicyclo[3.2.0]heptane-2-carboxylic acid (trihydrate)

CLASS Aminopenicillin (β-lactam antibiotic)

TRADE NAMES Amoxil · Ospamox · Moxypen · Flemoxin Solutab

THERAPEUTIC USES / INDICATIONS

- Acute otitis media, sinusitis, streptococcal pharyngitis.
- Community-acquired pneumonia (mild–moderate).
- Urinary tract infection (susceptible organisms), including in pregnancy.
- *H. pylori* eradication (with PPI + clarithromycin/metronidazole).
- Dental infections; endocarditis prophylaxis; Lyme disease.

ADMINISTRATION

- **PO:** capsules, tablets, suspension — with or without food.
- **IV** (some countries) by slow injection or infusion.
- Space doses evenly (time-dependent killing).
- Reconstituted suspension: refrigerate; discard after 14 days.

DOSAGES

- **Adult:** 500 mg every 8 h (or 875 mg every 12 h); severe: 1 g every 8 h.
- **Children:** 20–40 mg/kg/day divided every 8 h; otitis media high dose 80–90 mg/kg/day divided every 12 h.
- **CrCl <30 mL/min:** every 12 h; avoid 875 mg tablet.
- Usual course 5–7 days; streptococcal pharyngitis 10 days.

PHARMACOKINETICS

- **Peak:** 1–2 h. Bioavailability ~90%, unaffected by food.
- Widely distributed; crosses placenta; minimal liver metabolism.
- Excreted mainly unchanged in urine. **t½:** ~1 h (prolonged in renal failure).

PHARMACODYNAMICS

- Binds **penicillin-binding proteins (PBPs)** → inhibits cell-wall cross-linking → bacterial lysis (bactericidal).
- Time-dependent killing.
- Destroyed by **β-lactamases** (hence co-amoxiclav).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea.
- Mild maculopapular rash.
- Oral or vaginal thrush.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis (rare); urticaria, angioedema.
- Non-allergic rash in glandular fever (EBV).
- *C. difficile* colitis.
- Crystalluria with high IV doses.

CONTRAINDICATIONS & CAUTIONS

- Penicillin hypersensitivity.
- **Caution:** history of cephalosporin allergy, infectious mononucleosis, renal impairment.

KEY INTERACTIONS

- **Allopurinol:** ↑ rash.
- **Warfarin:** ↑ INR.
- **Methotrexate:** ↓ clearance → toxicity.
- **Live oral typhoid vaccine:** inactivated.

NURSING MANAGEMENT

- Ask about allergy before the first dose; record reaction type.
- Take cultures before first dose if ordered.
- Observe for 30 min after first dose; adrenaline available.
- Monitor temperature, WBC and clinical response.
- Report severe or bloody diarrhoea.

PATIENT EDUCATION

- Finish the whole course even if you feel better.
- Space doses evenly through the day.
- Stop and seek help for swelling, wheeze or widespread rash.
- Shake suspension well; keep in fridge.

KEY POINTS

- β-lactam; bactericidal; time-dependent.
- Rash in glandular fever is not a true allergy.
- Adult 500 mg every 8 h.
- Inactivated by β-lactamase → co-amoxiclav.